Huipeng Gong

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PROFILE

Builds and evaluates AI systems — LLM applications, retrieval pipelines, and the evaluation that establishes whether they work. Methodological training in the social sciences: structural equation modelling on panel data, computational text analysis, measurement validation. Moves between classical and computational methods as the research question demands, rather than committing to a single substantive topic.

EDUCATION

Durham University — United Kingdom Sep 2023 – Jan 2025
MSc in Sociology (English-taught programme) · Grade: Merit

Dissertation (advisor: Prof. Sol J. Gamsu): applied Bourdieu's capital theory and structural equation modelling in R to a **national panel dataset of 11,644 observations**, testing cultural capital as a mediator between socio-economic position and achievement.

Nantong University — China Sep 2018 – Jun 2023
BSc in International Economics and Trade · GPA: 85/100

Thesis: a time-series regression model in EViews on two decades of official trade data, estimating the effect of policy uncertainty on trade structure.

PUBLICATIONS

Xu, X., Wang, L., & **Gong, H.** "Digital Advocacy and Identity Performance: Multimodal Strategies of Environmental KOLs on Weibo." *Discourse, Context & Media* (SSCI), **under review**.

- Collected and cleaned large-scale platform data in Python; ran tokenisation, topic modelling and sentiment analysis; validated automated coding against a manually coded subsample with reported inter-coder reliability.

RESEARCH AND MODELLING PROJECTS

CCF Big Data & Computing Intelligence Contest · Network Analysis & Machine Learning Mar 2022 – Dec 2022
Ranked 9/159.

- Built multi-scale graph neural networks with attention to model complex networks at node, subgraph and global level.
- Added a noise-aware contrastive module re-weighting anchors to limit supervision error; designed a multi-dimensional ablation framework, improving accuracy by 18% over the baseline.

"Zhengda Cup" National Student Market Survey & Analysis Competition · Survey Research Dec 2020 – Jun 2021
3rd Prize, national.

- Led study design and instrument development; Likert-scale questionnaire with reliability and validity testing across multiple batches; multiple linear regression in SPSS.

PROFESSIONAL AND ENTREPRENEURIAL EXPERIENCE

Huawei Technologies · AI Application Development Engineer Oct 2025 – present

- Develop AI-driven application components in Python and C++, building data pipelines and evaluation benchmarks for multimodal human–AI interaction.
- Design multimodal HMI evaluation covering latency, robustness and edge cases, feeding model and interaction-experience tuning.

Droppulses · Founder (part-time venture, run alongside full-time employment) Jun 2025 – present

- Built and operate a system in which generative and multi-agent AI performs tasks autonomously, integrating large-language-model services into live commercial platforms via custom APIs; interaction logs provide ongoing empirical material on how AI-mediated exchange shapes user decisions.

Shanghai Value Enterprise Management Consulting · Data Analysis Intern Jun 2024 – Mar 2025

- Built an automated social-listening platform in Python with a MySQL architecture for real-time crawling, cleaning and classification; applied NLP to extract stakeholder concerns and sentiment at scale.

METHODS AND SKILLS

AI / LLM engineering — agent orchestration (ReAct-style loops, tool use); retrieval-augmented generation (hybrid keyword and dense retrieval, cross-encoder re-ranking); prompt and context engineering; evaluation design, baselines and rubric-based scoring.

Digital / computational — machine-learning classification; computational text analysis (NLP, topic modelling, sentiment analysis); network analysis (GNNs); web scraping and API-based data collection.

Classical social-science — structural equation modelling; mediation and moderation analysis; regression and econometric analysis; survey design and instrument validation; experimental design; multimodal qualitative coding.

Software — Python (Pandas, NumPy, scikit-learn, Flask), R, C++, SQL/MySQL, Git, Linux, SPSS, EViews, NVivo, Tableau.

Languages — Chinese (native); English (fluent, English-taught Master's).

HONOURS AND AWARDS

1st Prize, "Oriental Fortune Cup" Financial Challenge (2022) · 3rd Prize (national), "Zhengda Cup" Market Survey & Analysis Competition (2021) · 1st Prize, Student Innovation Training Competition, Nantong University (2021) · Honourable Mention, MCM/ICM (2020)

References available on request. · Latest version: gonghuipeng.com